

EC107: Microeconomics

In-class Exercises 3 — Elasticity & Its Application

Covers Day 4. For each **concept check**: vote, discuss with a neighbour, vote again. Short exercises are timed (5 minutes each).

A. Concept checks

- C1.** Which good most likely has the *most elastic* demand?
(a) insulin for a diabetic (b) salt
(c) one particular brand of cola (d) water in general
- C2.** A museum knows demand for entry is *inelastic*. To raise ticket revenue it should:
(a) lower the price (b) raise the price
(c) leave the price unchanged (d) impossible to say
- C3.** Moving down and to the right along a *straight-line* demand curve, demand becomes:
(a) more elastic (b) less elastic
(c) constant (equal to the slope) (d) perfectly elastic
- C4.** Demand for cigarettes is very inelastic. A new per-pack tax will be borne mostly by:
(a) sellers (b) smokers (c) split equally by law (d) the government

B. Short exercises

- N1. (Midpoint method & the TR test)** When the bus fare falls from £3 to £2, daily ridership rises from 800 to 1,200.
- (i) Compute $|\varepsilon_d|$ using the midpoint method.
 - (ii) Is demand elastic, inelastic, or unit elastic here?
 - (iii) Did total revenue rise or fall? Verify with the actual figures.
- N2. (Income & cross-price elasticities)**
- (i) A 10% rise in income raises a household's restaurant meals by 25%. Find the income elasticity, and classify the good (normal/inferior; luxury/necessity).
 - (ii) An 8% rise in the price of train tickets raises coach (bus) journeys by 4%. Find the cross-price elasticity, and state whether the goods are substitutes or complements.

C. Discussion

- D1.** A government introduces a petrol tax to cut consumption. Explain why the fall in consumption will be much larger in the *long run* than in the short run, using the idea of elasticity. What does this imply about evaluating the policy after only six months?